Weighted Local Linear Forecasting Under an El Niño Structural Break:
A Practical Extension of Cai–Gao–Selk with ML Benchmarks

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Abstract

We implement the Cai–Gao–Selk (CGS) weighted local linear (WLL) estimator in a real, event-driven financial setting: daily cocoa prices around an El Niño supply shock. We estimate the structural break nonparametrically, tune bandwidths and the combination weight γ with multifold forward validation (MFV), and compare WLL to Random Forest and XGBoost, including their pre/post convex combinations. WLL attains the lowest post-break MSFE in our runs and adapts as fast as ML methods in the early post-break window, while remaining interpretable.

1 Overview and headline findings

- Overall post-break OOS, WLL has the lowest MSFE; Pre-Break LL is a close second (Table 1).
- ML convex combos (RF/XGB) improve on single-regime RF/XGB but do not beat WLL.
- Early post-break, all models are close; WLL tracks the best ML closely (Figure 3).
- Modified DM tests rarely reject equality at 5%; some 10% evidence that WLL outperforms plain RF/XGB (Table 2).

2 Data, break, and tuning

We anchor the break on the El Niño event and also estimate it via Mohr–Selk (2020) on MFV-tuned local linear residuals. The detected and configured break dates are close, and volatility shifts after the break (Figure 1).

3 Models and MFV setup

- **NP layer**: Pre/Post LL (Epanechnikov), bandwidths by MFV; WLL combines them with γ by MFV on post-break data.
- **ML layer**: RF and XGB tuned by MFV; RF/XGB convex combos mirror WLL's γ logic.
- **Evaluation**: Post-break OOS MSFE; early-window cumulative squared error; modified DM for pairwise MSFE.

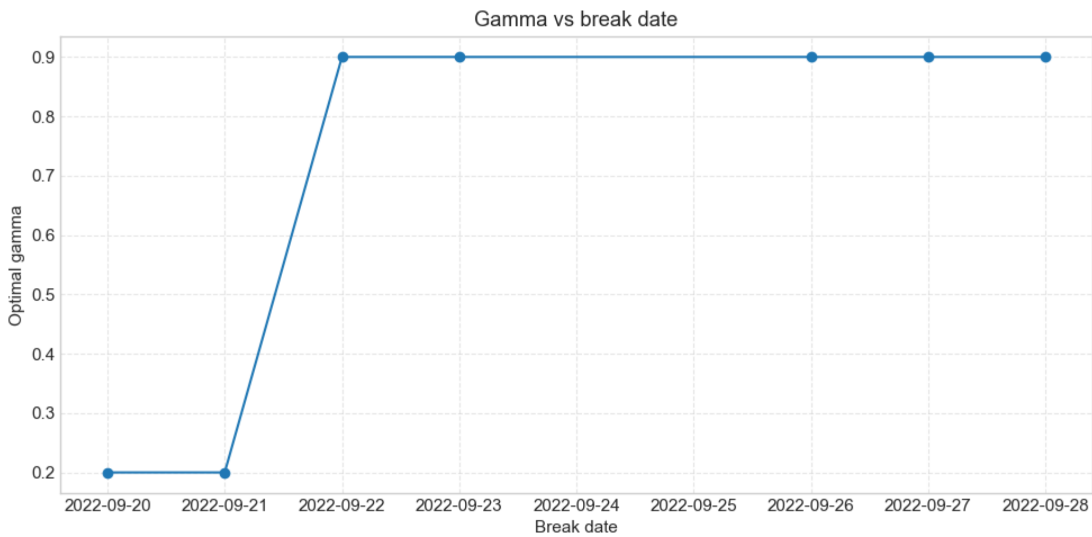

Figure 1: Detected vs configured break date and stability of γ around the El Niño event.

4 Results

4.1 MSFE rankings

Table 1: Post-break OOS MSFE (run: `msfe_results_20251201_191015.csv`)

Model	MSFE	RMSFE	Rank
WLL	0.0009425	0.03070	1
Pre-Break LL	0.0009429	0.03071	2
XGB Combo	0.0009501	0.03082	3
RF Combo	0.0010025	0.03166	4
Post-Break LL	0.0011006	0.03318	5
Random Forest	0.0012387	0.03520	6
XGBoost	0.0013043	0.03611	7

4.2 Early post-break adaptation

4.3 Modified DM tests

5 Interpretation: ML as NP under a break (funnel)

- RF/XGB are treated as nonparametric regressors; pre/post fits are combined with γ, same as WLL.
- WLL is the interpretable anchor; in this El Niño break it matches or beats ML on MSFE.
- Funnel: wide (flexible RF/XGB), middle (RF/XGB combos with γ), narrow (WLL). Here the funnel collapses toward WLL: ML does not dominate, and CGS-style combinations work for both kernels and trees.

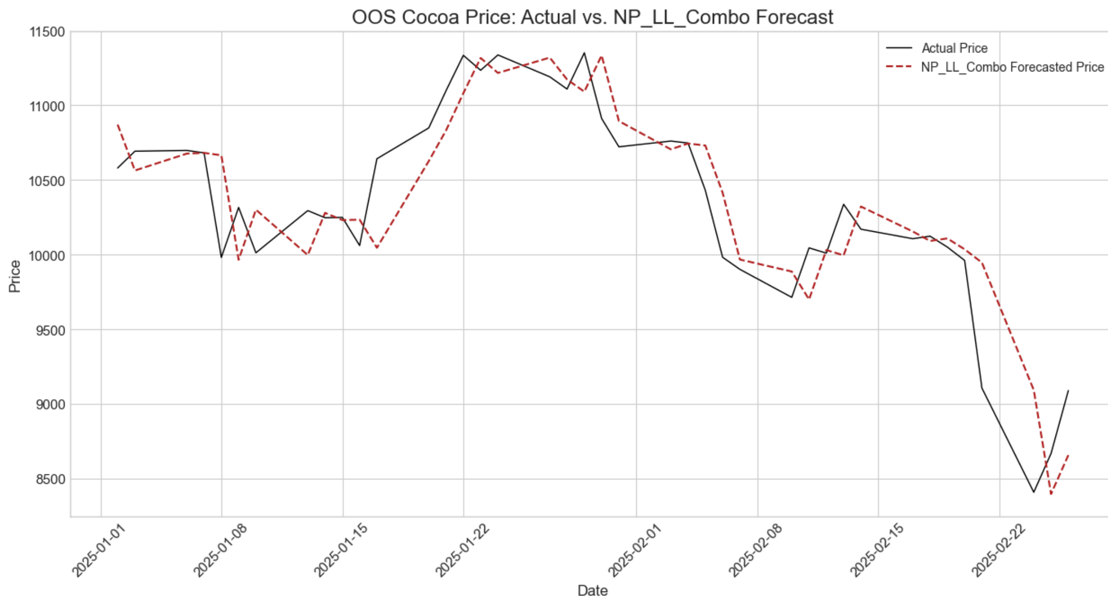

Figure 2: Post-break out-of-sample forecast trajectories across models.

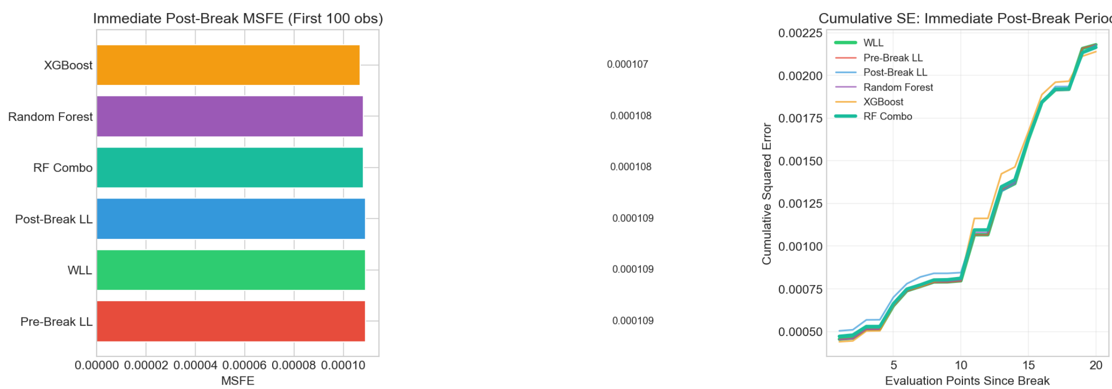

Figure 3: Early post-break cumulative squared error (first window after the break).

Table 2: Modified DM statistics (post-break OOS; `mdm_statistics_20251201_191015.csv`)

	PreLL	PostLL	WLL	RF	XGB	RFc	XGBc
PreLL	0.00	-1.22	0.06	-1.73	-1.93	-0.81	-0.34
PostLL	1.22	0.00	1.28	-3.22	-3.32	1.72	1.39
WLL	-0.06	-1.28	0.00	-1.80	-2.00	-0.89	-0.50
RF	1.73	3.22	1.80	0.00	-3.40	2.43	1.93
XGB	1.93	3.32	2.00	3.40	0.00	2.66	2.13
RFc	0.81	-1.72	0.89	-2.43	-2.66	0.00	1.00
XGBc	0.34	-1.39	0.50	-1.93	-2.13	-1.00	0.00